

Quiz (Carolina Reaper)

- Q1.** A cargo ship travels $240 \cdot 10^3$ meters in $4 \cdot 10^3$ seconds. What is its speed in meters per second?
(A) 50
(B) 60
(C) 70
(D) 80
(E) 90
- Q2.** A traffic signal has a cycle length of $2 \cdot 10^2$ seconds. If the green light lasts $8 \cdot 10$ seconds and the yellow light lasts $4 \cdot 10$ seconds, what fraction of the cycle is the red light?
(A) $\frac{1}{4}$
(B) $\frac{1}{3}$
(C) $\frac{3}{8}$
(D) $\frac{7}{16}$
(E) $\frac{3}{4}$
- Q3.** A company uses $5 \cdot 10^2$ liters of fuel to transport $3 \cdot 10^2$ crates. How many liters of fuel are needed to transport $12 \cdot 10^2$ crates?
(A) 1500
(B) 1600
(C) 2000
(D) 2500
(E) 3000
- Q4.** A delivery truck drives at 60 km/h for 2 hours and then at 40 km/h for 3 hours. What is the average speed of the truck?
(A) 45
(B) 48
(C) 50
(D) 52
(E) 55
- Q5.** The cost of shipping a package is directly proportional to its weight. If a 5 kg package costs 20 to ship, how much would a 12 kg package cost?
(A) 40
(B) 48
(C) 50
(D) 52
(E) 60
- Q6.** A bus travels 120 km in 2 hours. How far will it travel in 4.5 hours?
(A) 240
(B) 270
(C) 300
(D) 320
(E) 360
- Q7.** A ferry boat crosses a river at a rate of 2 km/min. If the river is 6 km wide, how many minutes does the crossing take?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 6
- Q8.** A train travels 800 km from City A to City B at an average speed of 80 km/h. How long does the trip take?
(A) 8 hours
(B) 10 hours
(C) 12 hours
(D) 14 hours
(E) 16 hours

Open Ended Questions (FRQ)

- Q9.** The distance d that a car travels is directly proportional to the time t it travels. If the car travels 240 km in 4 hours, find the equation of the line that represents this relationship and use it to find the distance the car travels in 6 hours.
- Q10.** A bicycle travels at a rate of 15 km/h for 2 hours and then at 20 km/h for 1.5 hours. Find the average speed of the bicycle for the entire trip and show your work.